

DAA-WEEK 7

CH.JYOTHIRMAI

CH.SC.U4CSE24107

JOB SEQUENCING:

CODE:

```
#include <stdio.h>
int jobs(int p[], int d[], int n) {
    for (int i = 0; i < n; i++) {
        int max = i;
        for (int j = i + 1; j < n; j++) {
            if (p[j] > p[max]) {
                max = j;
            }
        }
        int temp = p[max];
        p[max] = p[i];
        p[i] = temp;
        temp = d[max];
        d[max] = d[i];
        d[i] = temp;
    }
    int t[n + 1];
    for (int i = 0; i <= n; i++) {
        t[i] = 0;
    }
    int profit = 0;

    for (int i = 0; i < n; i++) {
        if (t[d[i]] == 0) {
            t[d[i]] = 1;
            profit += p[i];
        }
        else {
            continue;
        }
    }

    return profit;
}
```

```

int main() {
    int a;
    printf("Enter the number of profits: ");
    scanf("%d", &a);
    int p[a], d[a];
    for (int i = 0; i < a; i++) {
        printf("Enter the p[%d]: ", i);
        scanf("%d", &p[i]);
    }

    for (int i = 0; i < a; i++) {
        printf("Enter the d[%d]: ", i);
        scanf("%d", &d[i]);
    }

    int profits = jobs(p, d, a);

    printf("The total profit is: %d\n", profits);

    return 0;
}

```

OUTPUT:

```

Campus\SEM 4\DATA STRUCTURES\7 JOB
Enter the number of profits: 14
Enter the p[0]: 22
Enter the p[1]: 19
Enter the p[2]: 29
Enter the p[3]: 28
Enter the p[4]: 30
Enter the p[5]: 21
Enter the p[6]: 27
Enter the p[7]: 25
Enter the p[8]: 24
Enter the p[9]: 26
Enter the p[10]: 14
Enter the p[11]: 27
Enter the p[12]: 19
Enter the p[13]: 11
Enter the d[0]: 3
Enter the d[1]: 3
Enter the d[2]: 8
Enter the d[3]: 6
Enter the d[4]: 7
Enter the d[5]: 5
Enter the d[6]: 10
Enter the d[7]: 4
Enter the d[8]: 6
Enter the d[9]: 12
Enter the d[10]: 13
Enter the d[11]: 2
Enter the d[12]: 14
Enter the d[13]: 1
The total profit is: 279
PS C:\Users\ivoch\OneDrive - Amrita Vis

```